Hotel Booking Analytics

Business Intelligence & Machine Learning Report

Cancellation | Pricing | Seasonality | Customer | Market | ML Prediction

Executive Summary

This report presents a complete end-to-end data analysis and machine learning project on hotel booking data. It covers five business intelligence dimensions — cancellation behaviour, pricing dynamics, seasonality patterns, customer profiling, and market segmentation — followed by a predictive machine learning model for cancellation forecasting.

The dataset contains 33,798 cleaned bookings. The best performing model, XGBoost, achieved 85.47% accuracy and an AUC-ROC of 0.9135, with SHAP analysis confirming that lead time and number of special requests are the primary drivers of cancellation.

1. Key Business Metrics

1.1 Overall Performance

36K Total Bookings	**32.76%** Cancellation Rate	**$2.44M** Total Revenue
$99.93 Average Daily Rate	**$1.31M** Potential Revenue Loss	**3.02** Avg. Length of Stay

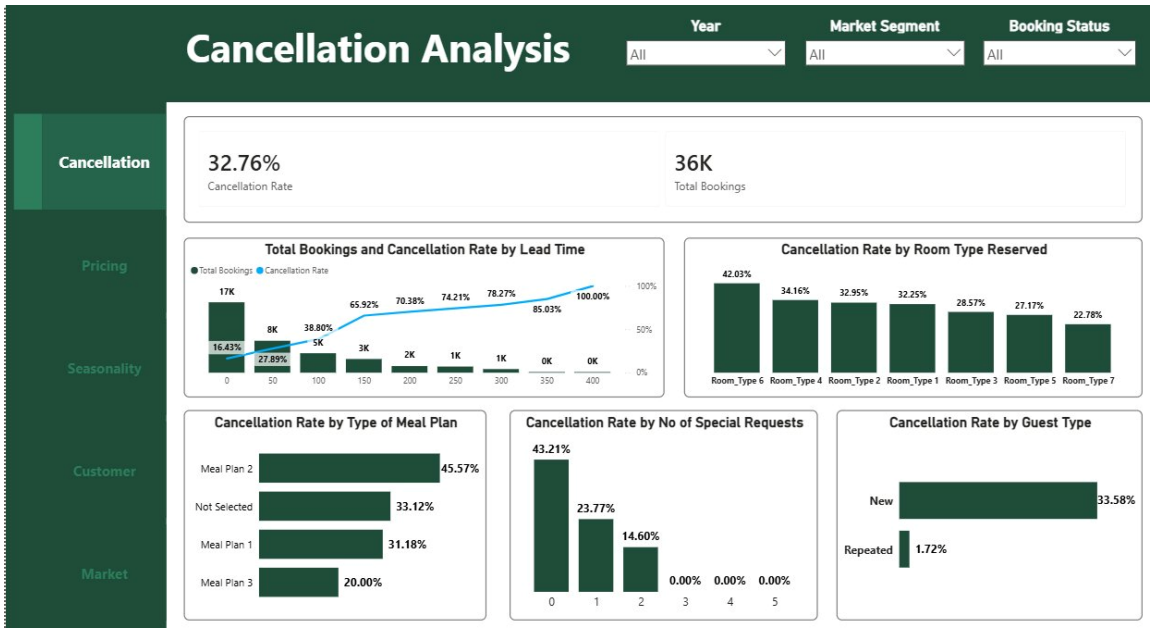
1.2 Seasonality & Customer KPIs

29K Weekend Nights	**80K** Weekday Nights
2.56% Repeated Guest Rate	**87 days** Avg. Lead Time (New Guests)

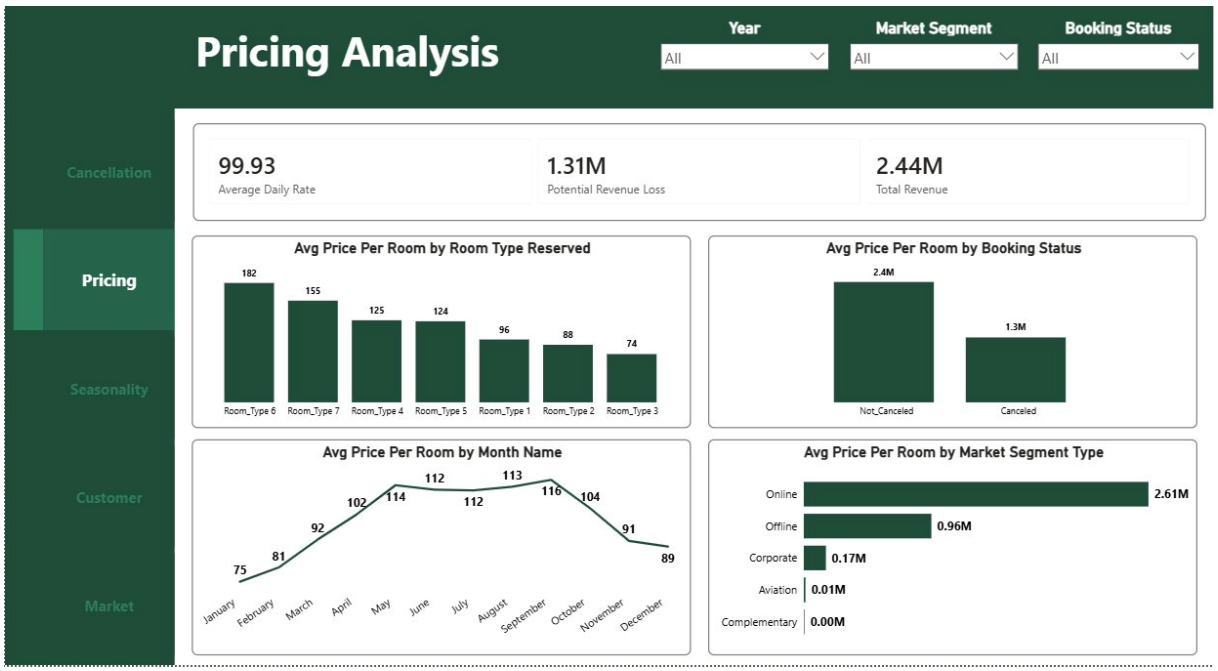
2. Dashboard Overview

The following pages from the interactive Power BI dashboard summarise key findings across all five analytical dimensions.

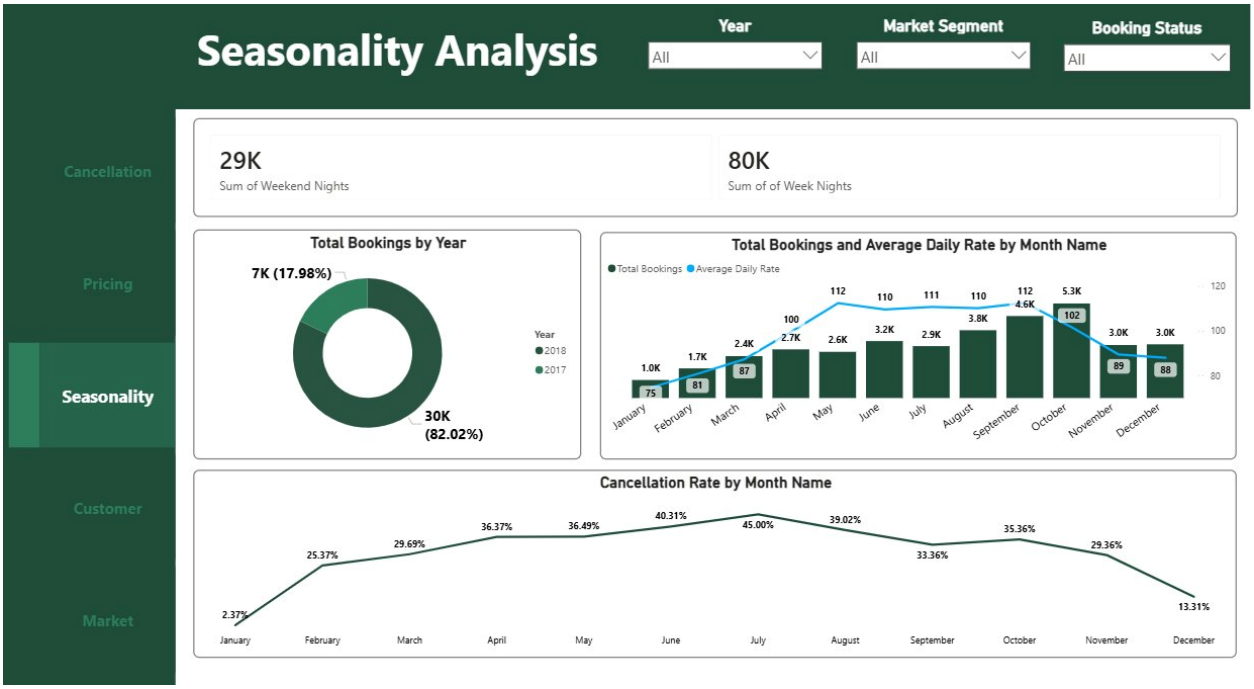
2.1 Cancellation Analysis



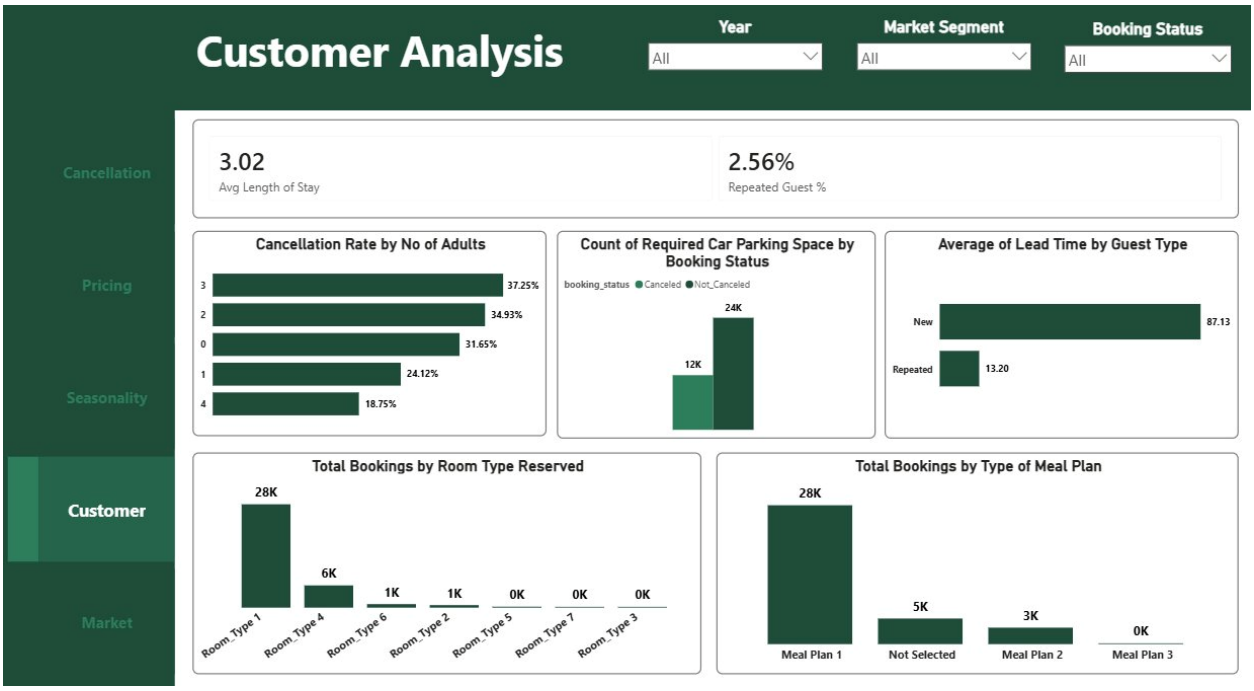
2.2 Pricing Analysis



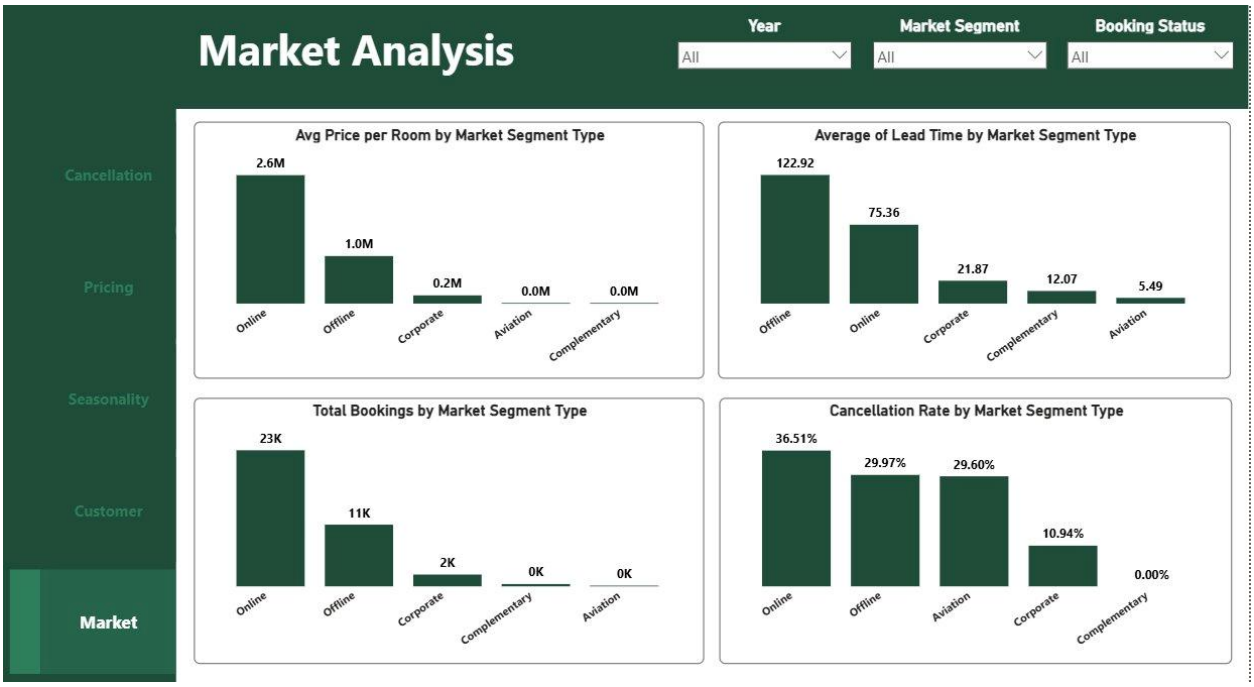
2.3 Seasonality Analysis



2.4 Customer Analysis



2.5 Market Analysis



3. Insights & Findings

3.1 Cancellation Analysis

High Overall Rate	At 32.76%, nearly 1 in 3 bookings is cancelled, representing $1.31M in potential lost revenue.
Lead Time Effect	Cancellation rates rise sharply with booking lead time: from 16.43% at 0 days to 100% at 350-400 days.
Room Type Risk	Room Type 6 has the highest cancellation rate (42.03%). Room Type 7 has the lowest at 22.78%.
Meal Plan Impact	Meal Plan 2 carries the highest cancellation risk at 45.57%, well above Meal Plan 3 at 20.00%.
Special Requests	Guests with 0 special requests cancel at 43.21%. Those with 2 requests cancel at only 14.60%.
Guest Type	New guests cancel at 33.58% while repeated guests cancel at just 1.72% — a 19x difference.

3.2 Pricing Analysis

Revenue Gap	Not-cancelled bookings generate $2.4M vs $1.3M for cancelled ones.
Room Premium	Room Type 6 commands a 2.5x price premium ($182) over the cheapest room type ($74).
Seasonal Peaks	Average daily rates peak in September-October ($113-$116). January is the lowest at $75.
Online Dominance	Online bookings account for $2.61M in total revenue vs $0.96M for offline.

3.3 Seasonality Analysis

Booking Volume	Volumes peak in October (5.3K) and September (4.6K). January is the quietest at 1K.
Summer Cancellations	Cancellation rates peak in July (45%) and June (40.31%) despite relatively high ADR.
Low-Season Commitment	January has only a 2.37% cancellation rate — late-bookers in low season are more committed.

Weekday Dominance	Weekday nights (80K) far outnumber weekend nights (29K).

3.4 Customer Analysis

Group Size Risk	3-adult bookings cancel most frequently (37.25%). Solo travellers cancel least at 24.12%.
Lead Time Gap	New guests book 87 days ahead on average; repeated guests book only 13 days ahead.
Room Preference	Room Type 1 dominates at 28K bookings. All other types are under 6K.
Meal Preference	Meal Plan 1 is overwhelmingly preferred at 28K bookings.

3.5 Market Analysis

Online Leads	Online segment has the largest volume (23K) and highest revenue ($2.61M) but 36.51% cancellation.
Corporate Reliability	Corporate segment has only 10.94% cancellation — the lowest among all segments.
Offline Stability	Offline segment (11K bookings) has lower cancellation (29.97%) than online.
Lead Time	Offline guests have the longest lead time at 122.92 days vs Online at 75.36 days.

4. Machine Learning Model

A binary classification model was built to predict whether a hotel booking will be cancelled. This allows the hotel to proactively identify high-risk bookings and take preventive action.

4.1 Data Preparation

Dataset	33,798 bookings after removing duplicates and cleaning
Features Used	15 features including lead_time, special_requests, market_segment, meal_plan, room_type, price
Target Variable	booking_status — Canceled (1) or Not_Canceled (0)
Class Balance	71.1% Not Canceled | 28.9% Canceled — reasonably balanced, no resampling needed
Train/Test Split	80% training | 20% testing with stratified sampling

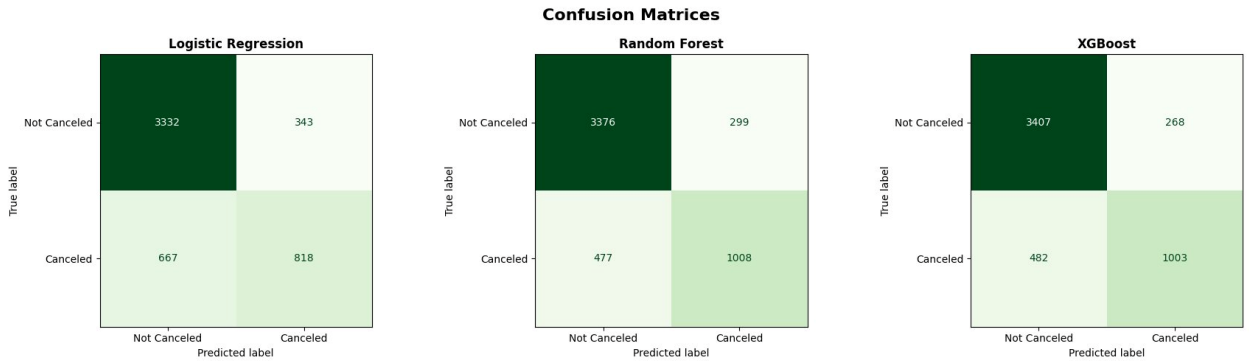
4.2 Model Comparison

Three models were trained and evaluated. XGBoost was selected as the final model with the highest AUC-ROC of 0.9135.

Model	Accuracy	Precision	Recall	F1 Score	AUC-ROC
Logistic Regression	80.43%	70.46%	55.08%	61.83%	0.8470
Random Forest	84.96%	77.12%	67.88%	72.21%	0.8992
XGBoost (Best)	**85.47%**	**78.91%**	**67.54%**	**72.79%**	**0.9135**

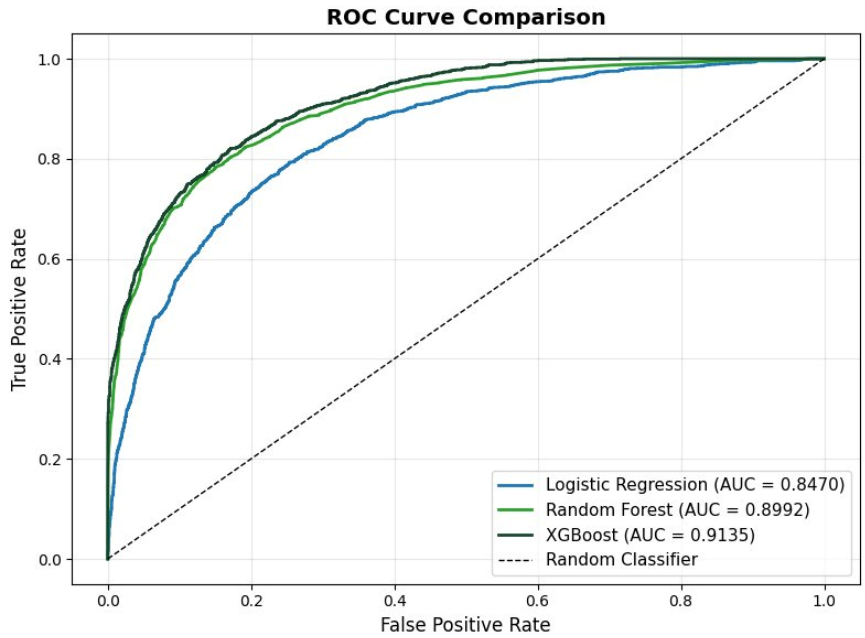
4.3 Confusion Matrices

The confusion matrices show how each model performs on the test set. XGBoost correctly identifies 3,407 non-cancellations and 1,003 cancellations with the fewest errors overall.



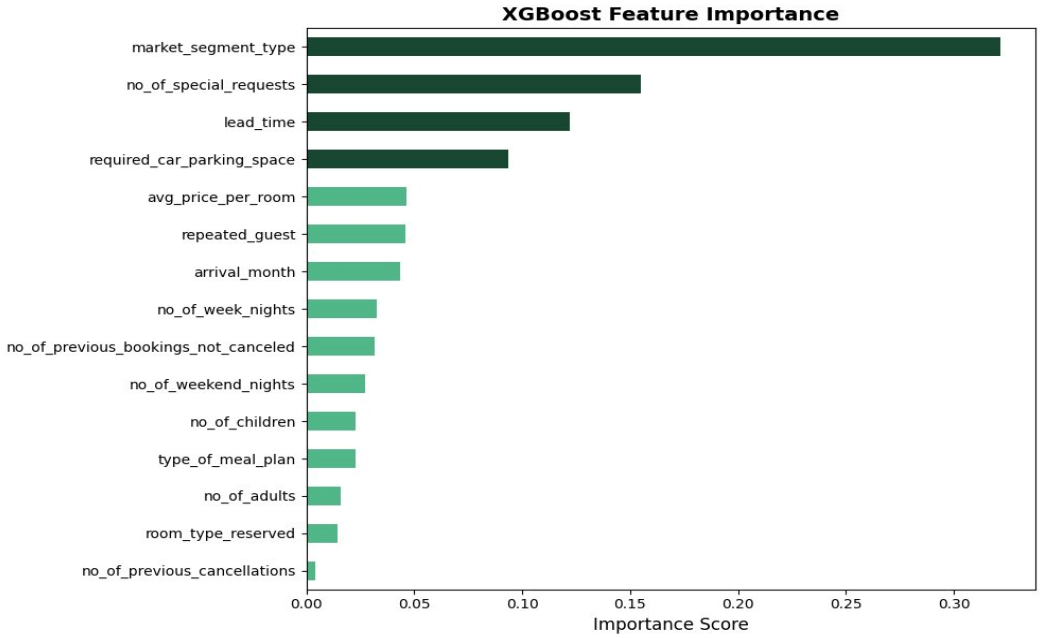
4.4 ROC Curve

The ROC curve compares all three models. XGBoost achieves the highest AUC of 0.9135 — correctly distinguishing between cancelled and non-cancelled bookings 91.35% of the time.



4.5 XGBoost Feature Importance

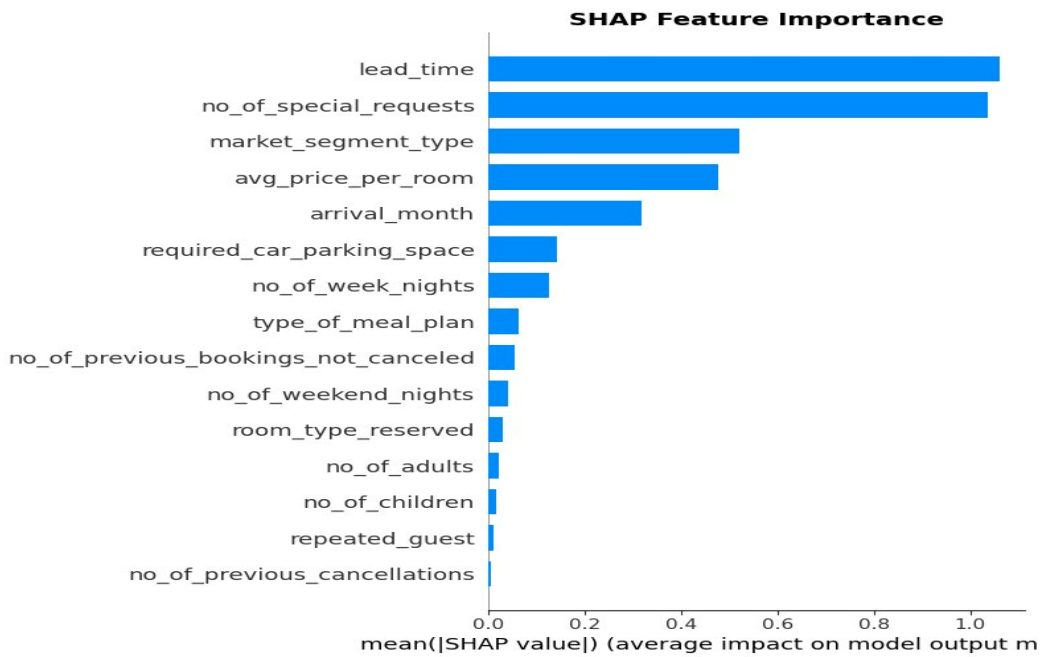
XGBoost built-in feature importance ranks market_segment_type first due to high cardinality. SHAP analysis below provides a more accurate and trustworthy ranking.



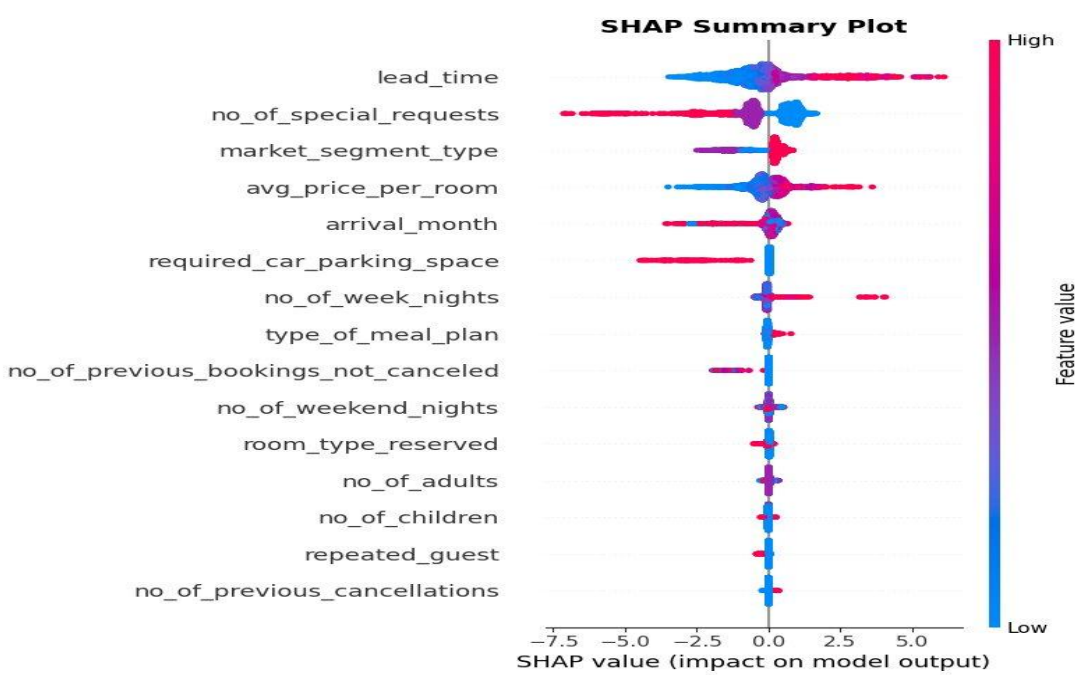
4.6 SHAP Explainability

SHAP (SHapley Additive exPlanations) measures the actual impact of each feature on individual predictions. It is more reliable than standard feature importance for business decision-making.

SHAP Feature Importance (Bar Chart) — average impact across all predictions:



SHAP Summary Plot — direction and magnitude for every individual booking:



4.7 Top Cancellation Drivers (SHAP)

Rank	Feature	Effect on Cancellation
1	lead_time	High lead time strongly increases cancellation probability
2	no_of_special_requests	More special requests strongly decrease cancellation probability
3	market_segment_type	Online segment increases cancellation risk
4	avg_price_per_room	Higher room price increases cancellation probability
5	arrival_month	Summer months (June-July) increase cancellation risk

4.8 Key ML Finding

An important discrepancy was found between XGBoost feature importance and SHAP rankings:

Feature Importance	Ranked market_segment_type as #1 — because it has many categories and appears in many tree splits
SHAP Analysis	Ranked lead_time as #1 and no_of_special_requests as #2 — reflecting true impact on predictions
Conclusion	SHAP is more trustworthy. Both methods confirm the patterns identified in the Power BI dashboard.

5. Business Recommendations

#	Area	Recommendation
1	**Cancellation Policy**	Introduce non-refundable rates for bookings made more than 90 days ahead where cancellation rates exceed 65%.
2	**Guest Retention**	Build a loyalty program — repeated guests cancel at 1.72% vs 33.58% for new guests, a 19x difference.
3	**Special Requests**	Proactively collect special requests at booking time. Each additional request significantly reduces cancellation risk.
4	**Meal Plan 2**	Audit Meal Plan 2 — its 45.57% cancellation rate suggests poor perceived value. Redesign or discontinue.
5	**Summer Policy**	June-July see peak ADR but 40-45% cancellation. Implement overbooking models calibrated to seasonal rates.
6	**Corporate Growth**	Expand B2B sales and corporate contracts — 10.94% cancellation is the lowest of any segment.
7	**Online Deposits**	Require deposits for online bookings with long lead times — they carry the highest cancellation risk.
8	**ML Deployment**	Deploy the XGBoost model to flag high-risk bookings in real time, enabling proactive intervention by staff.

6. Project Summary

Dimension	Key Metric	Risk Level	Top Action
Cancellation	32.76% rate	High	Deposit policy
Pricing	$99.93 ADR	Medium	Yield management
Seasonality	Peak: Oct/Sep	Medium	Dynamic pricing
Customer	2.56% repeat	High	Loyalty programme
Market	Online: 36.51%	High	Corporate growth
ML Model	AUC: 0.9135	Low	Deploy XGBoost